# Objectives

- ✓ Constructors

# Constructor

- ✓ A constructor **initializes an object immediately upon creation**. Once defined, the constructor is automatically called immediately after the object is created, before the new operator completes.
- ✓ **It has the same name as the class** in which it resides and is syntactically **like a method**.
- ✓ **Constructors have no return type**, not even void.

# Default Constructor

```
Box(){
        width  = 0;
        height = 0;
        depth  = 0;
}
```

**Default constructor Invocation**

- ✓ Object is initialized as specified in the default constructor

  *// declare, allocate, and initialize Box objects*
  ***Box mybox1 = new Box();***

- ✓ The Box( ) constructor assign default values for the variables

# Parameterized Constructor

A constructor that has parameters is known as **parameterized constructor**.

```
Box(double w, double h, double d)
{
width = w;
height = h;
depth = d;
}
```

- ✓ This is the constructor for Box-Parameterized constructor Invocation
- ✓ Object is initialized as specified in the parameters to its constructor.
- ✓ Declaration, allocation, and initialization of a Box object:-
  **Box mybox1 = new Box(10, 20, 15);**

# Method Vs Constructor

| Constructor | Method |
|---|---|
| Constructor is used to initialize the state of an object | Method is used to expose behaviour of an Object |
| Constructor must not have a return type | Method must have return type |
| Constructor is invoked implicitly | Method is invoked explicitly |
| Constructor name must be same as class name | Method name should not be same as class name |

# Constructor Overloading

Like method overloading constructor methods can be overloaded.

*Consider the constructor for the Box class as follows*

```
Box(double w, double h, double d){
width = w;
height = h;
depth = d;
}
```

**Whether *Box ob = new Box()* is valid??**

# Constructor Overloading

Overload the Box constructor for different requirements.

```
Box(double w, double h, double d){
        width = w;
        height = h;
        depth = d;
}

Box(double len){
        width = height = depth = len;
}

Box(){
        width = height = depth = -1;
}
```

# Constructor Overloading

- ✓ Various constructors for Box class.

```
Box mybox1 = new Box(10, 20, 15);
Box mybox2 = new Box();
Box mycube = new Box(7);
```

- ✓ Corresponding constructor will be invoked.

# Takeaways

- ✓ Constructors in java
- ✓ Initializing default constructor
- ✓ Parameterized constructor
- ✓ Overloading constructor

# Reference

[1] Herbert Schildt, *"Java: The Complete Reference, Eleventh Edition"*, *Oracle 2018*